

Recommendation Systems for Personalized Content Discovery

TEAM MEMBERS

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OBJECTIVE

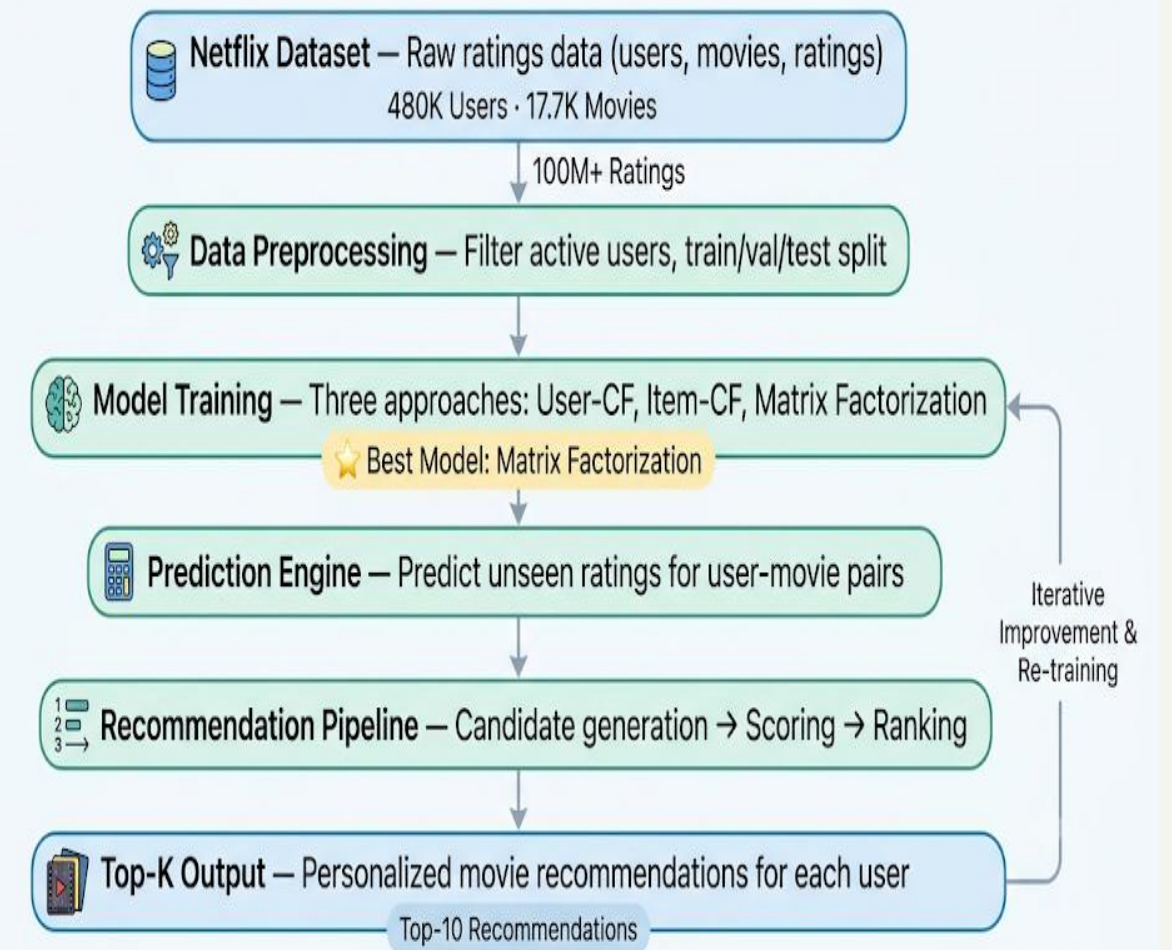
- Learn user preferences from historical ratings.
- Predict unseen movie ratings.
- Generate personalized Top-K recommendations.
- Compare collaborative filtering and matrix factorization methods.

DATASET

Netflix Prize Dataset

100.48M ratings | **480K** users | **17.7K** movies

Netflix Recommender System Workflow



Dataset & EDA Insights

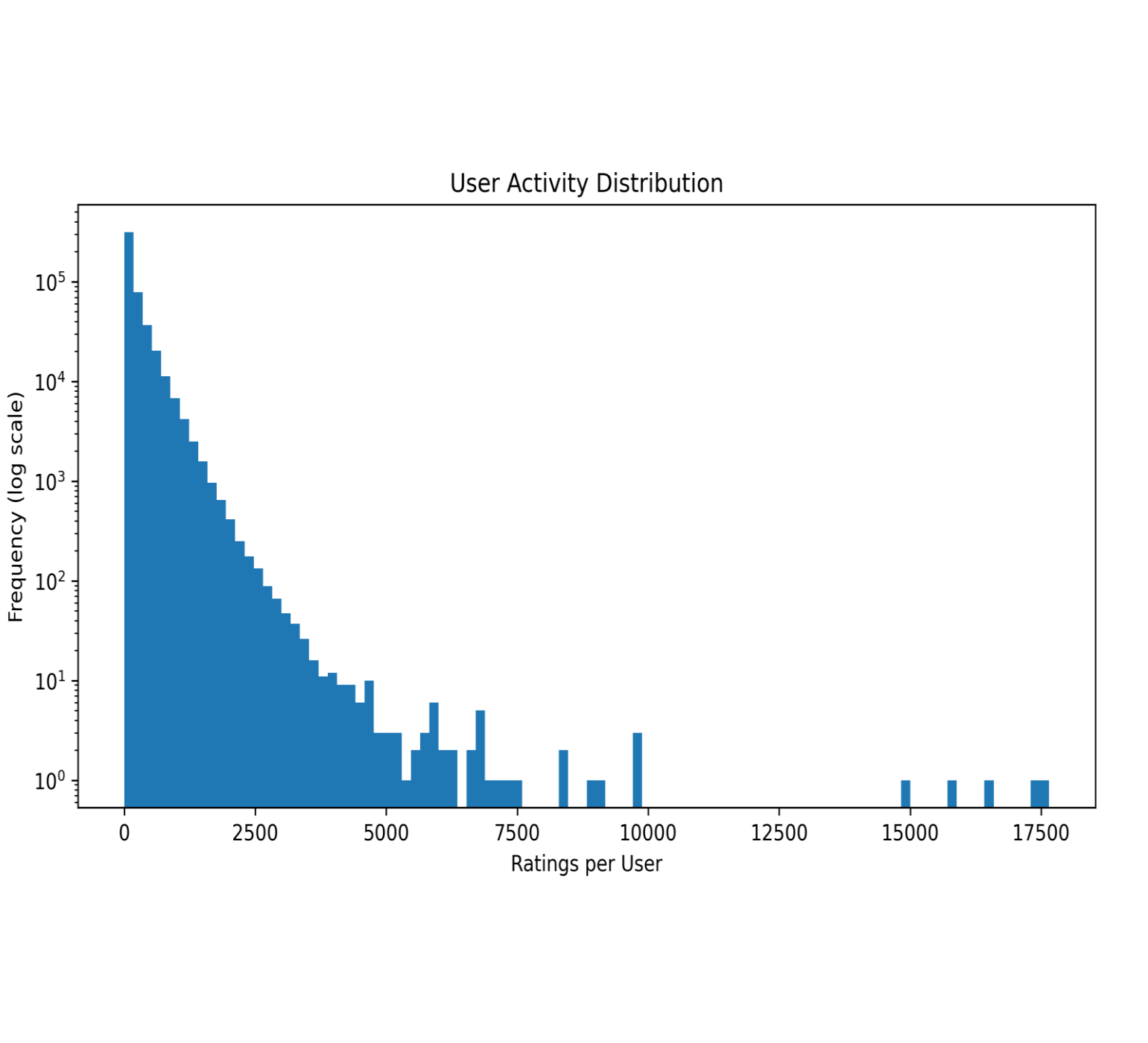
DATASET CHARACTERISTICS

Metric	Value
Users	480,189
Movies	17,770
Ratings	100,480,507
Sparsity	98.82%

KEY INSIGHTS

- User activity is highly skewed.
- Movie popularity follows a long-tail distribution.
- Ratings are concentrated around 3–5 stars.
- Extremely sparse user-item matrix.

Takeaway: The dataset is large, sparse, and highly imbalanced, making recommendation generation challenging.



Approach Overview

Baselines

- Global Mean
- User Mean
- Movie Mean

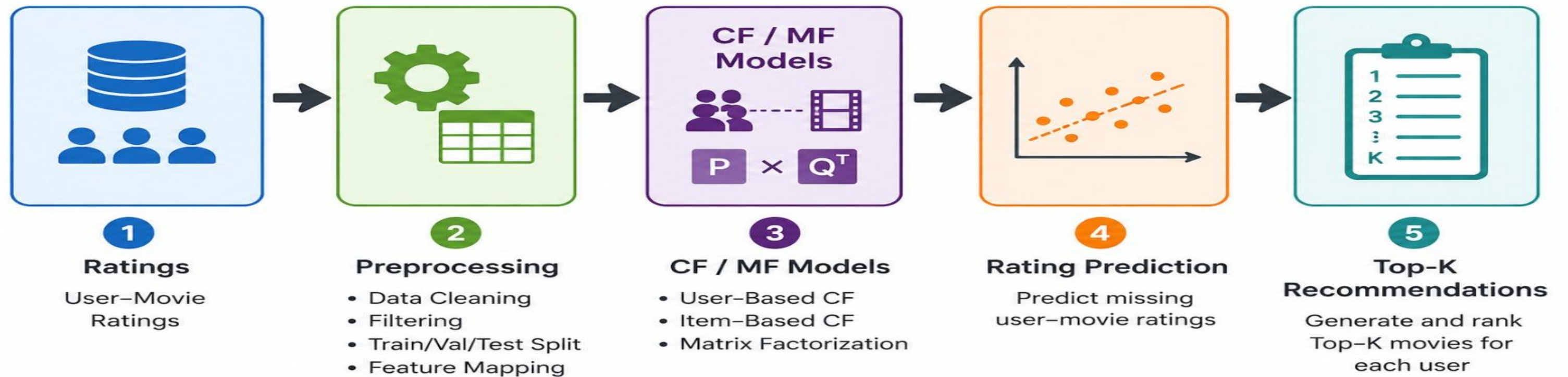
Collaborative Filtering

- User-Based CF
- Item-Based CF

Matrix Factorization

- Latent-factor recommendation model
- GPU accelerated PyTorch implementation

WORKFLOW



Matrix Factorization Model

PREDICTION FORMULA

$$\hat{r}_{ui} = \mu + b_u + b_i + p_u^T q_i$$

WHERE

- μ = global mean rating
- b_u = user bias term
- b_i = movie bias term
- p_u = user latent factor embedding
- q_i = movie latent factor embedding

Takeaway: Latent factors capture hidden relationships between users and movies.

TRAINING CONFIGURATION

Parameter	Value
Latent Factors	64
Optimizer	Adam
Batch Size	262,144
Epochs	15
Device	CPU

Evaluation Setup

TRAIN / VALIDATION / TEST SPLIT

Split	Ratio	Purpose
Train	80%	Model training
Validation	10%	Hyperparameter tuning
Test	10%	Final evaluation

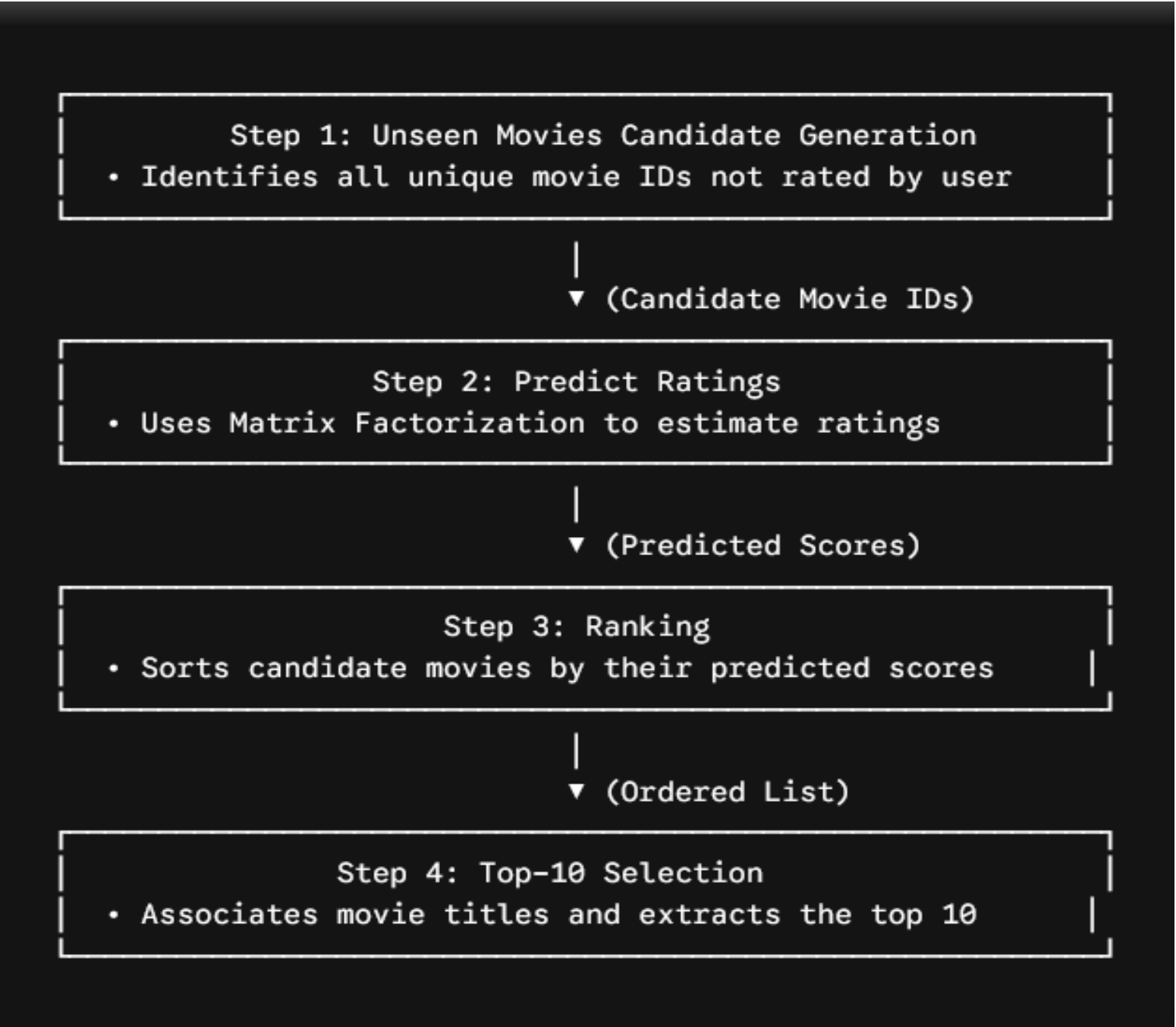
METRICS

RMSE — Measures rating prediction accuracy.

MAP@10 — Measures recommendation ranking quality.

Relevant movie: Rating ≥ 4

Takeaway: Both prediction accuracy and recommendation quality were evaluated.



Model Comparison Results

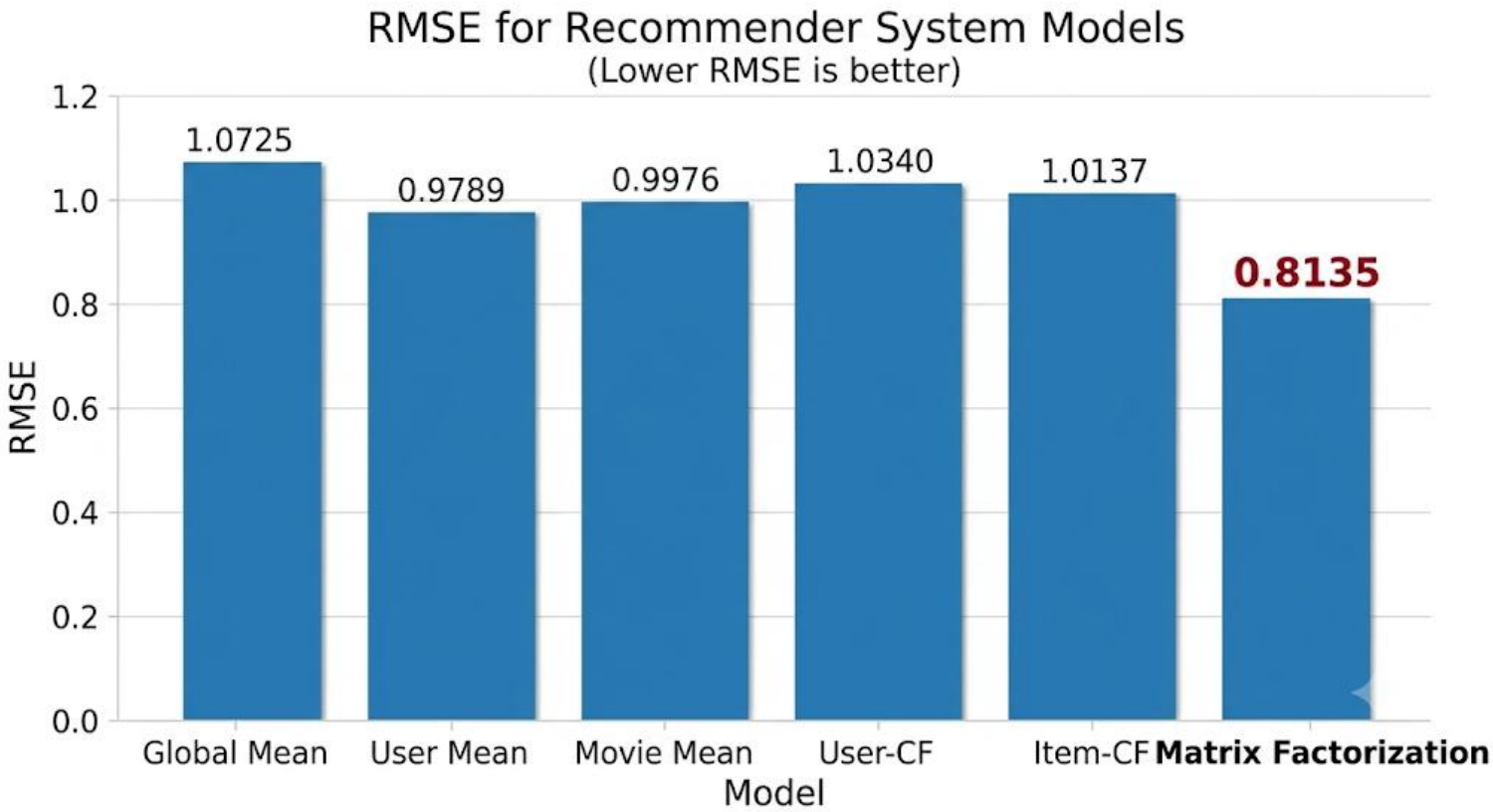
RMSE PERFORMANCE

Model	RMSE
Global Mean	1.0725
User Mean	0.9789
Movie Mean	0.9976
User-CF	1.0340
Item-CF	1.0137
Matrix Factorization	0.8135

ADDITIONAL METRICS

Metric	Value
MAP@10	0.04965
Coverage	9.45%

Takeaway: Matrix Factorization achieved the best overall performance.



Recommendation Examples

EXAMPLE USER

Previously Liked Movies	Recommended Movies
• Streets of Fire • Amadeus • Dangerous Liaisons • Moonlighting: The Pilot • The Ninth Gate	• Independence Day • The Silence of the Lambs • Pretty Woman • The Lord of the Rings: Fellowship of the Ring • The Green Mile

OBSERVATION

• Recommendations align with the user's historical interests. • The model successfully captures latent movie preferences.

Conclusions & Future Work

CONCLUSIONS

- Implemented and compared User-CF, Item-CF, and Matrix Factorization.
- Matrix Factorization achieved the best RMSE and recommendation quality.
- Successfully generated personalized movie recommendations.
- Demonstrated effectiveness of latent-factor models on large-scale sparse data.

KEY RESULTS

0.8135

RMSE

0.04965

MAP@10

9.45%

Coverage

FUTURE IMPROVEMENTS

- Hybrid recommendation systems.
- Cold-start handling.
- Content-based features.
- Ranking-aware loss functions.
- Diversity-aware recommendations.

FINAL TAKEAWAY

Matrix Factorization significantly outperformed neighborhood-based methods and proved most effective for personalized content discovery on the Netflix dataset.